

CHAPTER 11

Organisms and Populations

Major Biomes, Abiotic Factors and Responses to Abiotic Factors

- Niche is: (2018)
 - All the biological factors in the organism environment
 - The physical space where an organism live
 - The range of temperature that the organism needs to live
 - The functional role played by the organism where it lives
- Most animals are tree dwellers in a: (2015)
 - Temperature deciduous forest
 - Tropical rain forest
 - Coniferous forest
 - Thorn woodland
- Just as a person moving from Delhi to Shimla to escape the heat for the duration of hot summer, thousands of migratory birds from Siberia and other extremely cold northern regions move to: (2014)
 - Keolado National Park
 - Western Ghat
 - Meghalaya
 - Corbett National Park

Adaptations

- Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. (2023-Manipur)
Assertion (A): A person goes to high altitude and experiences "Altitude Sickness" with symptoms like breathing difficulty and heart palpitations.
Reason (R): Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.
 In the light of the above statements, choose the **correct** answer from the options given below:
 - (A) is true but (R) is false
 - (A) is false but (R) is true
 - Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).

- Match List-I with List-II:

List-I		List-II	
(A)	Allen's Rule	(i)	Kangaroo rat
(B)	Physiological adaptation	(ii)	Desert lizard
(C)	Behavioural adaptation	(iii)	Marine fish at depth
(D)	Biochemical adaptation	(iv)	Polar seal

Choose the correct answer from the options given below. (2021)

- A-iv B-i C-iii D-ii
 - A-iv B-i C-ii D-iii
 - A-iv B-iii C-ii D-i
 - A-iv B-ii C-iii D-i
- Assertion (A):** A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.
Reason (R): Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.
 In the light of the above statements, choose the correct answer from the options given below. (2021)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - (A) is false but (R) is true
 - Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Plants which produce characteristic pneumatophores and show vivipary belong to: (2017-Delhi)
 - Mesophytes
 - Halophytes
 - Psammophytes
 - Hydrophytes
 - It is much easier for a small animal to run uphill than for a large animal, because: (2016 - I)
 - It is easier to carry a small body weight
 - Smaller animals have a higher metabolic rate
 - Smaller animals have a lower O_2 requirement
 - The efficiency of muscles in large animals is less than in the small animals

9. Which of the following are likely to be present in deep sea water? (2014)

a. Saprophytic fungi
b. Archaeobacteria
c. Eubacteria
d. Blue-green algae

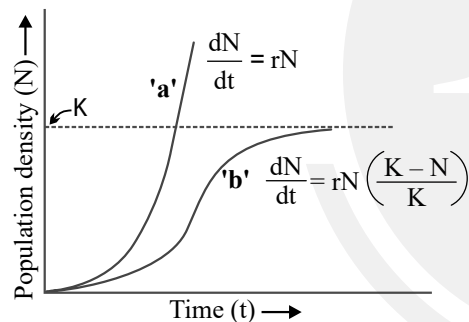
Population Attributes, Population Growth and Life History Variation

10. When will the population density increase, under special conditions? (2024 Re)

When the number of:

a. deaths exceeds number of births and also number of emigrants equals number of immigrants.
b. births plus number of immigrants equals number of deaths plus number of emigrants.
c. births plus number of emigrants is more than the number of deaths plus number of immigrants.
d. births plus number of immigrants is more than the sum of number of deaths and number of emigrants.

11. What do 'a' and 'b' represent in the following population growth curve? (2024 Re)



- a. 'a' represents exponential growth when responses are not limiting the growth; and 'b' represents logistic growth when responses are limiting the growth.
b. 'a' represents logistic growth when responses are not limiting the growth; 'b' represents exponential growth when responses are limiting the growth.
c. 'a' represents carrying capacity and 'b' shows logistic growth when responses are limiting the growth.
d. 'a' represents exponential growth when responses are not limiting the growth and 'b' shows carrying capacity.

12. The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$. (2024)

From this equation, K indicates:

a. Carrying capacity
b. Population density
c. Intrinsic rate of natural increase
d. Biotic potential

13. Match List-I with List-II. (2023)

List-I		List-II	
(A)	Logistic growth	(I)	Unlimited resource availability condition
(B)	Exponential growth	(II)	Limited resource availability condition
(C)	Expanding age pyramid	(III)	The percent individuals of pre-reproductive age is largest followed by reproductive and post reproductive age groups
(D)	Stable age pyramid	(IV)	The percent individuals of pre-reproductives and reproductive age group are same

Choose the correct answer from the options given below:

- a. A-II, B-IV, C-III, D-I b. A-II, B-I, C-III, D-IV
c. A-II, B-III, C-I, D-IV d. A-II, B-IV, C-I, D-III
14. If there are 250 snails in a pond, and within a year their number increases to 2500 by reproduction. (2023-Manipur)
What should be their birth rate per snail per year?
a. 10 b. 9 c. 25 d. 15
15. If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is _____ individuals per *Drosophila* per week. (2022)
a. Zero b. 0.1 c. 10 d. 1.0
16. In the exponential growth equation $N_t = N_0 e^{rt}$, e represents: (2021)
a. The base of exponential logarithms
b. The base of natural logarithms
c. The base of geometric logarithms
d. The base of number logarithms
17. Which of the following is not an attribute of a population? (2020)
a. Natality b. Mortality
c. Species interaction d. Sex ratio
18. The impact of immigration on population density is: (2020-Covid)
a. Both positive and negative
b. Neutralized by natality
c. Positive
d. Negative
19. Natality refers to: (2018)
a. Death rate
b. Birth rate
c. Number of individuals leaving the habitat
d. Number of individuals entering a habitat
20. In a growing population of a country: (2018)
a. Pre-reproductive individuals are more than the reproductive individuals.
b. Reproductive individuals are less than the post-reproductive individuals.
c. Reproductive and pre-reproductive individuals are equal in number.
d. Pre-reproductive individuals are less than the reproductive individuals.

21. Asymptote in a logistic growth curve is obtained when: (2017-Delhi)
- The value of 'r' approaches zero
 - $K = N$
 - $K > N$
 - $K < N$
22. When does the growth rate of a population following the logistic model equal zero? The logistic model is given as $\frac{dN}{dt} = rN(1 - N/K)$: (2016 - I)
- When N/K is exactly one
 - When N nears the carrying capacity of the habitat
 - When N/K equals zero
 - When death rate is greater than birth rate
23. Which of the following is correct for r-selected species? (2016 - II)
- Small number of progeny with small size
 - Small number of progeny with large size
 - Large number of progeny with small size
 - Large number of progeny with large size
24. A biologist studied the population of rats in a barn. He found that the average natality was 250, average mortality 240, immigration 20 and emigration 30. The net increase in population is: (2013)
- Zero
 - 10
 - 15
 - 05

Population Interactions

25. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Predator	(I)	<i>Ophrys</i>
(B)	Mutualism	(II)	<i>Pisaster</i>
(C)	Parasitism	(III)	Female wasp and fig
(D)	Sexual deceit	(IV)	<i>Plasmodium</i>

Choose the **correct** answer from the options given below:

- A-III, B-II, C-I, D-IV
 - A-IV, B-I, C-II, D-III
 - A-II, B-III, C-I, D-IV
 - A-II, B-III, C-IV, D-I
26. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Migratory flamingoes and resident fish in South American lakes	(I)	Interference competition
(B)	Abingdon tortoise became extinct after introduction of goats in their habitat	(II)	Competitive release
(C)	<i>Chathamalus</i> expands its distributional range in the absence of <i>Balanus</i>	(III)	Resource Partitioning
(D)	Five closely related species of Warblers feeding in different locations on same tree	(IV)	Interspecific competition

Choose the **correct** answer from the options given below:

- A-I, B-IV, C-III, D-II
 - A-IV, B-I, C-II, D-III
 - A-III, B-I, C-II, D-IV
 - A-II, B-IV, C-III, D-I
27. Given below are two statements :
- Statement I:** Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exists indefinitely.
- Statement II:** According to Gause's principle during competition, the inferior will be eliminated. This may be true if resources are limiting.
- In the light of the above statements, choose correct answer from the options given below: (2024)
- Statement I is true but Statement II is false.
 - Statement I is false but Statement II is true.
 - Both Statement I and Statement II are true.
 - Both Statement I and Statement II are false.
28. Given below are two statements:
- Statement I:** Gause's 'Competitive Exclusive Principle' states that two closely related species competing for the same related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually. (2023)
- Statement II:** In general, carnivores are more adversely affected by competition than herbivores.
- In the light of the above statements, choose the correct answer from the options given below:
- Statement I is incorrect but Statement II is true
 - Both Statement I and Statement II are true
 - Both Statement I and Statement II are false
 - Statement I is correct but Statement II is false.

29. Match List-I with List-II (2023)

List-I (Interaction)		List-II (Species A And B)	
(A)	Mutualism	(I)	+(A), O(B)
(B)	Commensalism	(II)	-(A), O(B)
(C)	Amensalism	(III)	+(A), -(B)
(D)	Parasitism	(IV)	+(A), +(B)

Choose the correct answer from the given options:

- A-III, B-I, C-IV, D-II
 - A-IV, B-II, C-I, D-III
 - A-IV, B-I, C-II, D-III
 - A-IV, B-III, C-I, D-II
30. Match List-I with List-II. (2023)

List-I (Interacting species)		List-II (Name of Interaction)	
(A)	A Leopard and a Lion in a forest/ grassland	(I)	Competition
(B)	A Cuckoo laying egg in a Crow's nest	(II)	Brood parasitism
(C)	Fungi and root of a higher plant in Mycorrhizae	(III)	Mutualism
(D)	A cattle egret and a Cattle in a field	(IV)	Commensalism

Choose the correct answer from the options given below:

- A-II, B-III, C-I, D-IV
- A-I, B-II, C-III, D-IV
- A-I, B-II, C-IV, D-III
- A-III, B-IV, C-I, D-II

31. Plants offer rewards to animals in the form of pollen and nectar and the animals facilitate the pollination process. This is an example of: (2023-Manipur)
 a. Amensalism b. Competition
 c. Commensalism d. Mutualism
32. The species of plants that plays a vital role in controlling the relative abundance of other species in a community is called (2023-Manipur)
 a. alien species b. endemic species
 c. exotic species d. keystone species
33. For chemical defence against herbivores, *Calotropis* has: (2023-Manipur)
 a. cardiac glycosides b. strychnine
 c. toxic ricin d. distasteful quinine
34. Two butterfly species are competing for the same nectar of a flower in a garden. To survive and coexist together, they may avoid competition in the same garden by (2022 Re)
 a. predating on each other
 b. feeding at the same time
 c. choosing different foraging patterns
 d. increasing time spent on attacking each other
35. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (–) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (–) for another species involved in the interaction? (2022)
 a. Competition b. Predation
 c. Amensalism d. Commensalism
36. Which one of the following statements **cannot** be connected to Predation? (2022)
 a. It is necessitated by nature to maintain the ecological balance
 b. It helps in maintaining species diversity in a community
 c. It might lead to extinction of a species
 d. Both the interacting species are negatively impacted
37. In spite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival? (2021)
 a. Competitive release b. Mutualism
 c. Predation d. Resource Partitioning
38. Amensalism can be represented as: (2021)
 a. Species A (+) ; Species B (+)
 b. Species A (–) ; Species B (–)
 c. Species A (+) ; Species B (0)
 d. Species A (–) ; Species B (0)
39. Match the items in Column-I with those in Column-II: (2020-Covid)

	Column - I		Column - II
1.	Herbivores-Plants	(i)	Commensalism
2.	Mycorrhiza-Plants	(ii)	Mutualism
3.	Sheep-Cattle	(iii)	Predation
4.	Orchid-Tree	(iv)	Competition

- | | | | | |
|----|-------|-------|-------|-------|
| | (1) | (2) | (3) | (4) |
| a. | (iii) | (ii) | (iv) | (i) |
| b. | (ii) | (i) | (iii) | (iv) |
| c. | (i) | (iii) | (iv) | (ii) |
| d. | (iv) | (ii) | (i) | (iii) |

40. Match Column - I with Column - II (2019)

	Column - I		Column - II
A.	Saprophyte	i.	Symbiotic association of fungi with plant roots
B.	Parasite	ii.	Decomposition of dead organic materials
C.	Lichens	iii.	Living on plants or animals
D.	Mycorrhiza	iv.	Symbiotic association of algae and fungi

Choose the correct answer from the option given below

- (A) (B) (C) (D)
 a. (i) (ii) (iii) (iv)
 b. (iii) (ii) (i) (iv)
 c. (ii) (i) (iii) (iv)
 d. (ii) (iii) (iv) (i)
41. Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other? (2018)
 a. *Hydrilla* b. *Yucca*
 c. Banana d. *Viola*
42. Which one of the following population interactions is widely used in medical science for the production of antibiotics? (2018)
 a. Commensalism b. Mutualism
 c. Parasitism d. Amensalism
43. Mycorrhizae are the example of: (2017-Delhi)
 a. Fungistasis b. Amensalism
 c. Antibiosis d. Mutualism
44. The principle of competitive exclusion was stated by: (2016 - II)
 a. MacArthur b. Verhulst and Pearl
 c. C.Darwin d. G.F. Gause
45. If '+' sign is assigned to beneficial interaction, '–' sign to detrimental and '0' sign to neutral interaction, then the population interaction represented by '+' '–' refers to: (2016 - II)
 a. Commensalism b. Parasitism
 c. Mutualism d. Amensalism
46. Gause's principle of competitive exclusion states that: (2016 - I)
 a. More abundant species will exclude the less abundant species through competition
 b. Competition for the same resources excludes species having different food preferences
 c. No two species can occupy the same niche indefinitely for the same limiting resources
 d. Larger organisms exclude smaller ones through competition

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|--|-------------------|---------------|--------------|----------------|--|---------------|-------------------|--------------|-----------------|
| <p>47. In which of the following interactions both partners are adversely affected?
(2015 Re)</p> <table border="0"> <tr> <td>a. Predation</td> <td>b. Parasitism</td> </tr> <tr> <td>c. Mutualism</td> <td>d. Competition</td> </tr> </table> | a. Predation | b. Parasitism | c. Mutualism | d. Competition | <p>48. A sedentary sea anemone gets attached to the shell lining of hermit crab. The association is:
(2013)</p> <table border="0"> <tr> <td>a. Amensalism</td> <td>b. Ectoparasitism</td> </tr> <tr> <td>c. Symbiosis</td> <td>d. Commensalism</td> </tr> </table> | a. Amensalism | b. Ectoparasitism | c. Symbiosis | d. Commensalism |
| a. Predation | b. Parasitism | | | | | | | | |
| c. Mutualism | d. Competition | | | | | | | | |
| a. Amensalism | b. Ectoparasitism | | | | | | | | |
| c. Symbiosis | d. Commensalism | | | | | | | | |

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (b) | 3. (a) | 4. (c) | 5. (b) | 6. (d) | 7. (b) | 8. (b) | 9. (b) | 10. (d) |
| 11. (a) | 12. (a) | 13. (b) | 14. (b) | 15. (b) | 16. (b) | 17. (c) | 18. (c) | 19. (b) | 20. (a) |
| 21. (b) | 22. (a) | 23. (c) | 24. (a) | 25. (d) | 26. (b) | 27. (b) | 28. (d) | 29. (c) | 30. (b) |
| 31. (d) | 32. (d) | 33. (a) | 34. (c) | 35. (b) | 36. (d) | 37. (d) | 38. (d) | 39. (a) | 40. (d) |
| 41. (b) | 42. (d) | 43. (d) | 44. (d) | 45. (b) | 46. (c) | 47. (d) | 48. (d) | | |

